MILITARY SPECIFICATION

ADAPTER SET, CLIP-IN SUBASSEMBLY ADU-18/C

1. SCOPE.- This specification covers equipment designated as Adapter Set, Clip-In Subassembly ADU-18/C, to be used as a component of the Clip-In Subassembly MHU-23/C.

2. APPLICABLE DOCUMENTS.- The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation; Methods Of
MIL-C-5015	Electrical Connectors
MIL-W-5088	Wiring; Aircraft Installation Of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-I-8500	Interchangeability and Replaceability, Physical, of Component Parts for Aircraft
MIL-T-9107	Test Reports, Preparation Of
MIL-C-13777	Cable, Special Purpose, Electrical
MIL-C-25200	Cable Assemblies, Special Weapons, Electrical, General Requirements For
MIL-S-25831	Support, Clip-In Subassembly MHU-13/C
MIL-C-25840	Clip-In Subassembly MHU-23/C
MIL-C-27002	Cartridge, Bomb Ejection MLU-2/C and Cartridge Kit, Bomb Ejection MLU-3/C

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS-33586	Metals - Definition of Dissimilar

FSC 1190

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PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

143 Specifications and Standards, Use Of

Air Force Specification Bulletin

23 Material and Process Specifications

Air Force Design Manual

80-1 Handbook of Instructions for Aircraft Designers

Index of Specifications and Related Publications (Used By) U. S. Air Force, Volume IV

DRAWINGS

Air Force

58J46480 Support, Clip-In Subassembly MHU-13/C
58J46484 Adapter Set, Clip-In Subassembly ADU-18/C

NOTE: All Air Force drawings listed in this specification and not listed in section 2 are subsidiary to 58J46484.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provision for preproduction testing.

3.2 MAJOR PARTS.- The ADU-18/C Adapter Set (58J46484) shall consist of the following major parts in the quantity indicated:

Parts	Quantity
Truss - Support, drawing 58J46597	2
Sway Brace, drawing 58J46594	1
Band Assembly, LH, drawing 58J46556	2
Band Assembly, RH, drawing 58J46460	2
Beam - Pullout, Arming Cable, drawing 58J46500	1
Torque Tube Assembly, drawing 58J46537	1
Cable Assembly, Cartridge, drawing 58D46643	1

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Parts	Quantity
Cable Assembly, Release, drawing 58D46644	1
Cartridge Kit, Bomb Ejection MLU-3/C (MIL-C-27002)	2
Cartridge, Bomb Ejection MLU-2/C (MIL-C-27002)(2 per Kit)	4
Release Assembly, drawing 58J46506, including the following expendable parts:	1
Nut - Explosive Release, drawing 58J46547	2
Stud, drawing 58D46638	1
Sleeve, drawing 58D46551	2
Washer Assembly, drawing 58C46546	2

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143, except as provided in 3.3.1 and 3.3.2; and provided that, for materials and processes, the applicable group I items per ANA Bulletin 143 shall be selected first from USAF Specification Bulletin No. 23 and, if not included therein, then from the Index of Specifications and Related Publications (Used By), U. S. Air Force, Volume IV.

3.3.1 COMMERCIAL PARTS.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts like screws, bolts, nuts, cotter pins, etc., having suitable properties may be used provided:

a. They can be replaced by the standard parts (AN or MS) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 STANDARD PARTS.- Except for the use authorized in 3.3.1, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 MATERIAL.- Materials shall be selected and processed for the various components in accordance with the applicable specifications and standards as indicated on the drawings.

3.5 DESIGN AND CONSTRUCTION.- The ADU-18/C Adapter Set shall be designed and constructed in accordance with drawing 58J46484.

3.5.1 DISSIMILAR METALS IN CONTACT.- Dissimilar metals in contact as defined in MS-33586 shall be protected from corrosion in accordance with the applicable provisions and references in Air Force Design Manual 80-1 (HIAD), Volume I.

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3.6 PERFORMANCE

3.6.1 PERFORMANCE OF ADU-18/C ADAPTER SET.- The ADU-18/C, when properly assembled with an MHU-13/C Support and mounted on a Rack, Clip-In System MAU-6/A, shall perform in accordance with the requirements of MIL-C-25840.

3.6.2 PERFORMANCE OF INDIVIDUAL COMPONENTS

3.6.2.1 CARTRIDGE KIT, BOMB EJECTION MLU-3/C.- The MLU-3/C Cartridge Kit shall conform to the requirements of MIL-C-27002.

3.6.2.2 CARTRIDGE, BOMB EJECTION MLU-2/C.- The MLU-2/C Cartridge shall conform to the requirements of MIL-C-27002.

3.6.2.3 CABLE ASSEMBLY, CARTRIDGE.- The cable assembly shall conform to the requirements of MIL-C-13777, MIL-C-25200 and drawing 58D46643.

3.6.2.4 CABLE ASSEMBLY, RELEASE.- The cable assembly shall conform to the requirements of MIL-C-13777, MIL-C-25200, and drawing 58D46644.

3.7 ELECTRICAL REQUIREMENTS

3.7.1 ELECTRICAL CONNECTORS.- Electrical connectors shall be in accordance with MIL-C-5015.

3.7.2 ELECTRICAL WIRING.- Electrical wiring shall be in accordance with MIL-W-5088.

3.7.3 ELECTRICAL WIRE NUMBERS.- Electrical wire numbers shall utilize "AY" and "TZA" prefixes as specified on the drawings.

3.7.4 The adapter set shall fulfill all the operating requirements specified herein when operating under any voltage from 18 to 30 volts dc.

3.7.5 EXPLOSION PROOF.- All electrical components shall be explosion proof.

3.8 INTERCHANGEABILITY.- All adapter sets shall be interchangeable. All components shall be replaceable, as defined in MIL-I-8500. All permanently joined members shall be interchangeable or replaceable in their joined state. (See 6.5)

3.9 DIMENSIONS AND TOLERANCES.- Dimensions and tolerances shall conform to the applicable drawings.

3.10 WEIGHT.- The weight of the adapter set shall not exceed 230 pounds.

3.11 IDENTIFICATION OF PRODUCT.- Identification shall be in accordance with MIL-STD-130 except for the requirements of 3.11.1.

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3.11.1 NAMEPLATE.- The nameplate furnished with the adapter set shall show identification for the Clip-In Subassembly MHU-23/C in accordance with drawing 58B46652.

3.12 WORKMANSHIP.- The adapter set, including all parts and accessories, shall be fabricated in a manner that will insure that the criteria of appearance, fit, and adherence to specified tolerances are observed. Particular attention shall be given to neatness and thorough ness of marking of parts and assemblies, plating, painting, riveting and welding. All parts shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the ADU-18/C Adapter Set shall be as follows:

a. Preproduction Tests
b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES.- The preproduction test articles shall be representative of production adapter sets. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 PREPRODUCTION TEST REPORTS.- In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS.- Preproduction tests shall consist of all tests described herein under "4.4 TEST METHODS."

4.3 ACCEPTANCE TESTS.- Acceptance tests shall consist of the following individual tests as described herein under "4.4 TEST METHODS":

a. Examination of Product
b. Conformance to Drawings
c. The Functional Tests of MIL-C-25840.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- Each component shall be given a complete visual examination to determine compliance with the requirements of this specification, including workmanship.

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4.4.2 CONFORMANCE TO DRAWINGS.- Each component shall be inspected for strict conformance to the engineering drawings and all specifications and standards referenced therein.

4.4.3 ELECTRICAL COMPONENTS.- All electrical components shall be subjected to explosion-proof tests in accordance with procedure III of MIL-E-5272, except that fuel used shall be equivalent to, or more flammable than, the optimum mixture of JP-4 fuel and air. Commercial butane or propane is a satisfactory fuel for this purpose.

4.4.4 TESTING IN ACCORDANCE WITH MIL-C-25840.- The ADU-18/C Adapter Set shall be assembled with an MHU-13/C Support, (MIL-S-25831, drawing 58J46480), and mounted on an MAU-6/A Rack, and shall then be tested in accordance with all preproduction testing provisions of MIL-C-25840.

4.4.5 BASIS FOR REJECTION

a. Individual components failing to meet their respective tests shall be rejected. Rejected components either shall be replaced with satisfactory units or shall be reworked to correct the defects, after which all specified tests shall be repeated.

   Rejected components shall not be resubmitted for inspection or test without furnishing full particulars concerning previous rejections and measures taken to overcome defects.

b. All parts, specimens, or assemblies destroyed in making tests required by this specification shall be in addition to the quantity specified in the order and shall be furnished to the procuring activity without increasing the cost of the contract. Accepted units shall have such acceptance marking on the unit as specified by the procuring activity.

c. After any test, the occurrence of any of the following conditions shall be sufficient cause for rejection.

   Failure to perform within required tolerances.

   Failure of structure, whether by deformation, rupture, or misalignment.

   Dirty or pitted electrical contact points.

   Dirt, metal chips, filings, solder, or burrs on any surface.

   Parts not properly finished.

   Loose rivets, screws, pins or bolts.

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5. PREPARATION FOR DELIVERY

5.1 MLU-3/C CARTRIDGE KIT.- The MLU-3/C Cartridge Kit shall be prepared for delivery in accordance with MIL-C-27002.

5.2 ADAPTER SET, LESS MLU-3/C CARTRIDGE KIT

5.2.1 PRESERVATION.- Preservation for all shipment levels shall be in accordance with method I of MIL-P-116.

5.2.2 PACKAGING AND PACKING.- The quantity of adapter sets per container shall be specified by the procuring activity.

5.2.2.1 LEVELS A AND B.- For level A and B shipment, ADU-18/C Adapter Sets shall be physically protected in accordance with JAN-P-100 and packed in a box conforming to PPP-B-621.

5.2.2.2 LEVEL C.- The ADU-18/C Adapter Set for level C shipment shall be packed in exterior type containers in such a manner as to insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules, or the regulations of other common carriers as applicable.

5.2.3 MARKING FOR SHIPMENT AND STORAGE.- Interior packages and exterior shipping containers shall be marked for identification in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

ADAPTER SET, CLIP-IN SUBASSEMBLY ADU-18/C

6. NOTES

6.1 INTENDED USE.- The ADU-18/C Adapter Set, when assembled with an MHU-13/C Support, will form a Clip-In Subassembly MHU-23/C intended for carrying a weapon, internally, in high-performance bomber aircraft.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Required levels of preservation, packaging and packing.

c. Whether preproduction testing will be conducted at a laboratory designated by the procuring activity or by the contractor under the supervision of the procuring activity.

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6.3 ITEMS WHICH MAY BE PROCURED SEPARATELY.- If considered advantageous to the Government, the following items may be contracted for separately:

a. Cartridge Kit, Bomb Ejection MLU-3/C.

b. Cable Assembly, Cartridge, Bomb Ejection, Single Store.

c. Cable Assembly, Release, Single Store.

If the listed items are procured separately, they need comply only with their respective specifications and drawings.

6.4 PREPRODUCTION TEST SAMPLES.- It is expected that the contract or purchase order will specify that 2 complete ADU-18/A Adapter Sets will be required as first articles, and that these first articles will be subjected to the preproduction tests to determine compliance with this specification.

6.5 DEFINITIONS

a. PREPRODUCTION TESTS.- Preproduction tests are those tests performed prior to production on ADU-18/C Adapter Sets representative of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS.- Acceptance tests are those tests accomplished on equipment submitted for acceptance under contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction items.

c. PERMANENTLY JOINED MEMBERS.- Members shall be regarded as permanently joined if they are staked, swaged, welded, soldered, riveted, or pressed-fit, or if they are aligned or built to close tolerances.

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